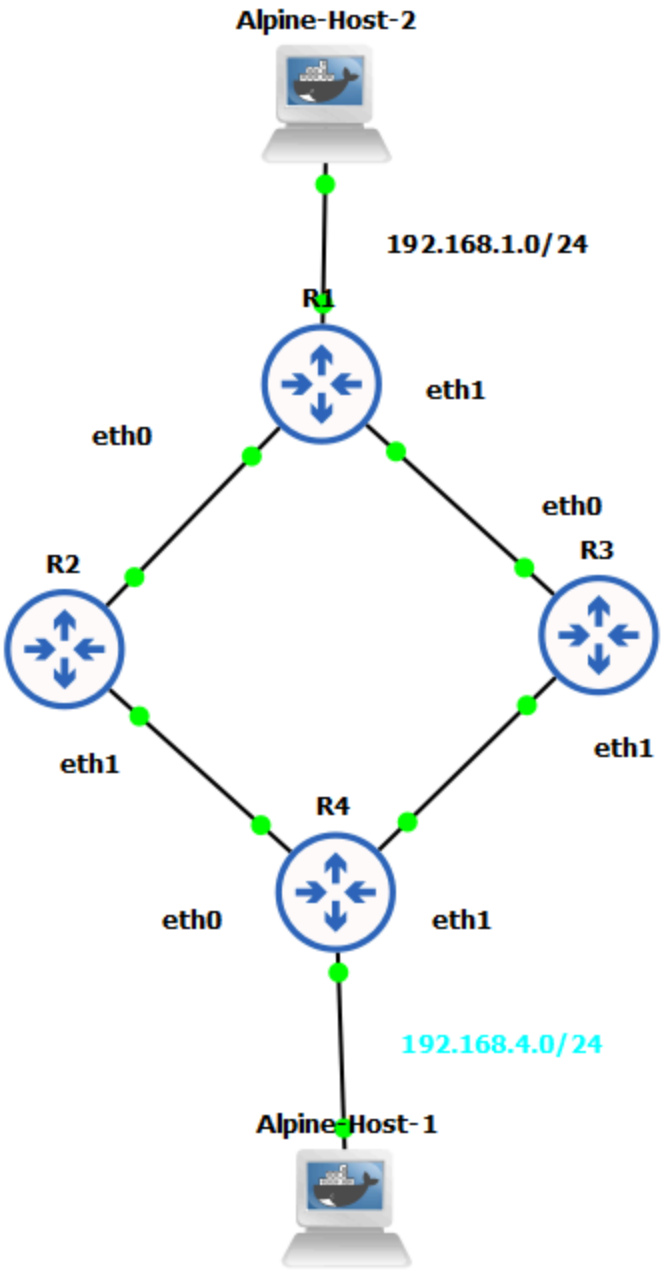
Network topology.



Phase 3 :

Step 1

```
Host 1 ip set up
ip addr add 192.168.4.10/24 dev eth0
ip link set eth0 up
ip route add default via 192.168.4.1

`Host 2 set up`
ip addr add 192.168.1.10/24 dev eth0
ip link set eth0 up
ip route add default via 192.168.1.1
```

I'm pinging the host on the R1 side the host 2 with the ping having the date time printed so i can determine the timeframe of how quick or how long does the 2nd path takes over

```
   ping 192.168.1.10 | while read l; do echo "$(date +%T) $l"; done
```

to do it i suspended the link between r1 and r2 and closely monitoring the ping and here's what i got

“Pasted image 20261102103142.png” could not be found.

it took about 3 seconds and 2 pings we're missing as clearly shown in the sequence number Seq 296 and 297 are missing, so **2 pings were lost**.

Number for your table: by our formula (first reply after the gap minus last reply before it) that is 02:30:03 minus 02:30:00 = **3 seconds**, with 2 packets lost. If you got 4 by counting differently, tell me how, but use 3 for consistency in all tests. Ping only has 1-second precision, so say “about 3 seconds”.

this is from the monitor log command on R1

“Pasted image 20261102103214.png” could not be found.

the flaw in gns3 is that it somehow notifies the router that the wire has been pulled and as R1 felt it instantly and dropped the neighbor without waiting for the **40 seconds dead timer**

Now collecting proof that the failover really did work and without a doubt being used by R1

here's some proof

from R1

“Pasted image 20261102104743.png” could not be found.

notice the line 10.0.0.12/30 is reachable via 2 paths
O> 10.0.0.12/30 [110/40] via 10.0.0.2, eth0 inactive, weight 1, 00:11:20

- `via 10.0.0.6, eth1, weight 1, 00:11:20`

10.0.0.2, eth0 inactive, weight 1, 00:11:20 this one is the path via R2 and since the link between the routers are suspended therefore this one is inactive which serves as a **proof that the 2nd path to reach R4's lan via r3 is now working** proven by this line **via 10.0.0.6, eth1, weight 1, 00:11:20** .

and to back it up

“Pasted image 20261102105623.png” could not be found.

the image shows that it uses the path via r3 instead of r2

1st hop was the r4 lan default gateway
2nd hop was the r3's eth1 interface form r4
3rd hop was from r3 to the eth1 ip address of r1
4th was to the host we are pinging

Question arising

```
Does traffic return to the main path with the lowest cost by itself  when the
link is not suspended anymore and how long does it take ?
```

now the link is back monitor log indicates the adjacency being formed again
**

“Pasted image 20261102115044.png” could not be found.

and the traceroute command proves That **YES** the primary path is used again

---

Test 3 Lose the whole router

```
What i did was  to lose the router 2 entirely so i press stop on it to
simulate it shutting down
```

upon shutting it down the pings seems to stop for a while meaning packets are being dropped.

as shown here

```
ping 192.168.1.10 | while read l; do echo "$(date +%T) $l"; done
```

```
05:47:42 PING 192.168.1.10 (192.168.1.10): 56 data bytes
05:47:42 64 bytes from 192.168.1.10: seq=0 ttl=61 time=2.429 ms
05:47:43 64 bytes from 192.168.1.10: seq=1 ttl=61 time=2.044 ms
05:47:44 64 bytes from 192.168.1.10: seq=2 ttl=61 time=2.275 ms
05:47:45 64 bytes from 192.168.1.10: seq=3 ttl=61 time=2.204 ms
05:47:46 64 bytes from 192.168.1.10: seq=4 ttl=61 time=2.136 ms
05:47:47 64 bytes from 192.168.1.10: seq=5 ttl=61 time=1.836 ms
05:47:48 64 bytes from 192.168.1.10: seq=6 ttl=61 time=2.315 ms
05:47:49 64 bytes from 192.168.1.10: seq=7 ttl=61 time=2.096 ms
05:47:50 64 bytes from 192.168.1.10: seq=8 ttl=61 time=1.940 ms
05:47:51 64 bytes from 192.168.1.10: seq=9 ttl=61 time=2.318 ms
05:47:52 64 bytes from 192.168.1.10: seq=10 ttl=61 time=2.278 ms
05:47:53 64 bytes from 192.168.1.10: seq=11 ttl=61 time=2.225 ms
05:47:54 64 bytes from 192.168.1.10: seq=12 ttl=61 time=1.908 ms
05:47:55 64 bytes from 192.168.1.10: seq=13 ttl=61 time=1.871 ms
05:47:56 64 bytes from 192.168.1.10: seq=14 ttl=61 time=1.920 ms
05:47:57 64 bytes from 192.168.1.10: seq=15 ttl=61 time=2.066 ms
05:47:58 64 bytes from 192.168.1.10: seq=16 ttl=61 time=2.778 ms
05:47:59 64 bytes from 192.168.1.10: seq=17 ttl=61 time=1.789 ms
05:48:00 64 bytes from 192.168.1.10: seq=18 ttl=61 time=1.895 ms
05:48:01 64 bytes from 192.168.1.10: seq=19 ttl=61 time=1.887 ms
05:48:02 64 bytes from 192.168.1.10: seq=20 ttl=61 time=1.840 ms
05:48:03 64 bytes from 192.168.1.10: seq=21 ttl=61 time=2.553 ms
05:48:04 64 bytes from 192.168.1.10: seq=22 ttl=61 time=1.959 ms
05:48:05 64 bytes from 192.168.1.10: seq=23 ttl=61 time=3.011 ms
05:48:06 64 bytes from 192.168.1.10: seq=24 ttl=61 time=2.147 ms
05:48:07 64 bytes from 192.168.1.10: seq=25 ttl=61 time=2.378 ms
05:48:08 64 bytes from 192.168.1.10: seq=26 ttl=61 time=2.520 ms
05:48:09 64 bytes from 192.168.1.10: seq=27 ttl=61 time=2.431 ms
05:48:10 64 bytes from 192.168.1.10: seq=28 ttl=61 time=2.458 ms
05:48:11 64 bytes from 192.168.1.10: seq=29 ttl=61 time=2.404 ms
05:48:12 64 bytes from 192.168.1.10: seq=30 ttl=61 time=2.231 ms
05:48:13 64 bytes from 192.168.1.10: seq=31 ttl=61 time=2.260 ms
05:48:14 64 bytes from 192.168.1.10: seq=32 ttl=61 time=1.800 ms
05:48:48 64 bytes from 192.168.1.10: seq=66 ttl=61 time=2.481 ms
05:48:49 64 bytes from 192.168.1.10: seq=67 ttl=61 time=1.860 ms
05:48:50 64 bytes from 192.168.1.10: seq=68 ttl=61 time=2.000 ms
05:48:51 64 bytes from 192.168.1.10: seq=69 ttl=61 time=2.120 ms
05:48:52 64 bytes from 192.168.1.10: seq=70 ttl=61 time=2.129 ms
```

```
05:48:53 64 bytes from 192.168.1.10: seq=71 ttl=61 time=2.420 ms
05:48:54 64 bytes from 192.168.1.10: seq=72 ttl=61 time=1.984 ms
05:48:55 64 bytes from 192.168.1.10: seq=73 ttl=61 time=1.900 ms
05:48:56 64 bytes from 192.168.1.10: seq=74 ttl=61 time=1.861 ms
05:48:57 64 bytes from 192.168.1.10: seq=75 ttl=61 time=2.060 ms
05:48:58 64 bytes from 192.168.1.10: seq=76 ttl=61 time=2.289 ms
05:48:59 64 bytes from 192.168.1.10: seq=77 ttl=61 time=2.307 ms
05:49:00 64 bytes from 192.168.1.10: seq=78 ttl=61 time=2.279 ms
05:49:01 64 bytes from 192.168.1.10: seq=79 ttl=61 time=1.746 ms
```

This shows the **ping gap** the last reply 5:48:14 (seq 32) then the first reply after that is 5:48:48 (seq 66) so the outage is 48 - 14 = 34 seconds with 33 pings lost (seq 33 to 65).

R1's log

```
vyos@R1:~$ monitor log
Nov 03 05:38:45 systemd[1]: systemd-tmpfiles-clean.service: Deactivated
successfully.
Nov 03 05:38:45 systemd[1]: Finished Cleanup of Temporary Directories.
Nov 03 05:38:45 systemd[1]: run-credentials-
systemd\x2dtmpfiles\x2dclean.service.mount: Deactivated successfully.
Nov 03 05:39:53 kernel: e1000: eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow
Control: RX
Nov 03 05:39:53 vyos-netlinkd[2375]: RTM_NEWLINK -> eth0, state=UP,
mac=0c:49:97:01:00:00
Nov 03 05:39:54 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on
eth0:10.0.0.1: Down -> Init (HelloReceived)
Nov 03 05:40:03 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on
eth0:10.0.0.1: Init -> ExStart (2-WayReceived)
Nov 03 05:40:03 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on
eth0:10.0.0.1: ExStart -> Exchange (NegotiationDone)
Nov 03 05:40:03 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on
eth0:10.0.0.1: Exchange -> Full (ExchangeDone)
Nov 03 05:44:11 rsyslogd[1978]: -- MARK --
Nov 03 05:48:54 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 ==🟠(default) on
eth0:10.0.0.1: Full -> Deleted (InactivityTimer)==
```

R1 log: only one line this time: `Full -> Deleted (InactivityTimer)`. There is **no "NIC Link is Down"** line, unlike Test 1

Route :

```
vyos@R1:~$ show ip route ospf
Codes: K - kernel route, C - connected, L - local, S - static,
R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
```

```
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric, t - Table-Direct,
```

```
  - selected route, * - FIB route, q - queued, r - rejected, b - backup
```

```
t - trapped, o - offload failure
```

```
IPv4 unicast VRF default:
O>* 3.3.3.3/32 [110/20] via 10.0.0.6, eth1, weight 1, 00:26:57
O>* 4.4.4.4/32 [110/40] via 10.0.0.6, eth1, weight 1, 00:02:41
O>* 10.0.0.8/30 [110/50] via 10.0.0.6, eth1, weight 1, 00:02:34
O>* 10.0.0.12/30 [110/40] via 10.0.0.6, eth1, weight 1, 00:02:41
O>* 192.168.4.0/24 [110/41] via 10.0.0.6, eth1, weight 1, 00:02:41
```

This shows the command output show ip route ospf is correct 192.168.4.0/24 goes via **10.0.0.6** (R3), cost 41, and 2.2.2.2 is gone.

To confirm the path taken with traceroute

/ # traceroute 192.168.1.10
traceroute to 192.168.1.10 (192.168.1.10), 30 hops max, 46 byte packets
1 192.168.4.1 (192.168.4.1) 0.776 ms 0.461 ms 0.485 ms
2 10.0.0.13 (10.0.0.13) 1.546 ms 1.810 ms 1.627 ms
3 10.0.0.5 (10.0.0.5) 1.661 ms 2.017 ms 1.428 ms
4 192.168.1.10 (192.168.1.10) 1.640 ms 2.424 ms 1.552 ms

192.168.4.1, then 10.0.0.13, then 10.0.0.5, then the host. This is the backup path.

```
RESULT: outage about 34 seconds, 33 pings lost.
```

```
Detection: OSPF Dead timer (InactivityTimer). No link-down message, because a
stopped router does not report the cable as down.
```

```
Path after: R1 - R3 - R4 (backup).
```

```
Note: Host-1 and R1 clocks differ by a few seconds, so only Host-1 ping times are
used for the outage.
```

One thing to know: the R1 log says 05:48:54, but your pings were already back at 05:48:48. That cannot happen on the same clock, so the Host-1 clock and the R1 clock are a few seconds apart. Your 34 seconds is correct because it uses only Host-1's times. Never subtract a time from the router log and a time from the ping. Write this in the file as a note.

Test 4 Suspend R3-R4 (the backup road) while everything else is healthy

does a broken backup road cause any visible outage?

to test this i suspended the link between r4 and r3

r4's log

```
vyos@R4:~$ monitor log
Nov 03 06:32:08 (systemd)[2519]: pam_unix(systemd-user:session): session opened
for user vyos(uid=1002) by (uid=0)
Nov 03 06:32:08 systemd[2519]: Queued start job for default target
default.target.
Nov 03 06:32:08 systemd[2519]: Reached target paths.target - Paths.
Nov 03 06:32:08 systemd[2519]: Reached target sockets.target - Sockets.
Nov 03 06:32:08 systemd[2519]: Reached target timers.target - Timers.
Nov 03 06:32:08 systemd[2519]: Reached target basic.target - Basic System.
Nov 03 06:32:08 systemd[1]: Started User Manager for UID 1002.
Nov 03 06:32:08 systemd[2519]: Reached target default.target - Main User Target.
Nov 03 06:32:08 systemd[2519]: Startup finished in 143ms.
Nov 03 06:32:08 systemd[1]: Started Session 3 of User vyos.
Nov 03 06:34:03 kernel: e1000: eth1 NIC Link is Down
Nov 03 06:34:03 ospfd[1372]: AdjChg: Nbr 3.3.3.3, NbrIP 10.0.0.13 (default) on
eth1:10.0.0.14: Full -> Deleted (KillNbr)
Nov 03 06:34:03 vyos-netlinkd[2391]: RTM_NEWLINK -> eth1, state=DOWN,
mac=0c:a8:7b:35:00:01
```

R4's log verifies that the link between R3 and R4 is suspended
and it also shows that the adjacency is dropped between the two routers

---

To support that claim even further

R4's ospf neigbor

```
vyos@R4:~$ show ip ospf neighbor

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
2.2.2.2 1 Full/- 20m05s 34.595s 10.0.0.9 eth0:10.0.0.10 0 0 0
```

---

ping from host 1 to 192.168.1.10 shows no packet is dropped during the outage of the backup path

```
/ # ping 192.168.1.10 | while read l; do echo "$(date +%T) $l"; done
06:33:43 PING 192.168.1.10 (192.168.1.10): 56 data bytes
06:33:43 64 bytes from 192.168.1.10: seq=0 ttl=61 time=3.439 ms
06:33:44 64 bytes from 192.168.1.10: seq=1 ttl=61 time=2.362 ms
06:33:45 64 bytes from 192.168.1.10: seq=2 ttl=61 time=1.879 ms
06:33:46 64 bytes from 192.168.1.10: seq=3 ttl=61 time=2.095 ms
06:33:47 64 bytes from 192.168.1.10: seq=4 ttl=61 time=2.207 ms
06:33:48 64 bytes from 192.168.1.10: seq=5 ttl=61 time=1.934 ms
06:33:49 64 bytes from 192.168.1.10: seq=6 ttl=61 time=2.324 ms
06:33:50 64 bytes from 192.168.1.10: seq=7 ttl=61 time=2.665 ms
06:33:51 64 bytes from 192.168.1.10: seq=8 ttl=61 time=2.198 ms
06:33:52 64 bytes from 192.168.1.10: seq=9 ttl=61 time=2.566 ms
06:33:53 64 bytes from 192.168.1.10: seq=10 ttl=61 time=1.759 ms
06:33:54 64 bytes from 192.168.1.10: seq=11 ttl=61 time=1.828 ms
06:33:55 64 bytes from 192.168.1.10: seq=12 ttl=61 time=2.124 ms
06:33:56 64 bytes from 192.168.1.10: seq=13 ttl=61 time=2.099 ms
06:33:57 64 bytes from 192.168.1.10: seq=14 ttl=61 time=2.089 ms
06:33:58 64 bytes from 192.168.1.10: seq=15 ttl=61 time=2.354 ms
06:33:59 64 bytes from 192.168.1.10: seq=16 ttl=61 time=2.184 ms
06:34:00 64 bytes from 192.168.1.10: seq=17 ttl=61 time=1.903 ms
06:34:01 64 bytes from 192.168.1.10: seq=18 ttl=61 time=2.858 ms
06:34:02 64 bytes from 192.168.1.10: seq=19 ttl=61 time=2.027 ms
06:34:03 64 bytes from 192.168.1.10: seq=20 ttl=61 time=2.252 ms
06:34:04 64 bytes from 192.168.1.10: seq=21 ttl=61 time=2.133 ms
06:34:05 64 bytes from 192.168.1.10: seq=22 ttl=61 time=2.113 ms
06:34:06 64 bytes from 192.168.1.10: seq=23 ttl=61 time=2.018 ms
06:34:07 64 bytes from 192.168.1.10: seq=24 ttl=61 time=2.093 ms
06:34:08 64 bytes from 192.168.1.10: seq=25 ttl=61 time=2.077 ms
06:34:09 64 bytes from 192.168.1.10: seq=26 ttl=61 time=2.377 ms
06:34:10 64 bytes from 192.168.1.10: seq=27 ttl=61 time=2.895 ms
06:34:11 64 bytes from 192.168.1.10: seq=28 ttl=61 time=1.846 ms
06:34:12 64 bytes from 192.168.1.10: seq=29 ttl=61 time=1.998 ms
06:34:13 64 bytes from 192.168.1.10: seq=30 ttl=61 time=2.691 ms
06:34:14 64 bytes from 192.168.1.10: seq=31 ttl=61 time=2.007 ms
06:34:15 64 bytes from 192.168.1.10: seq=32 ttl=61 time=1.921 ms
06:34:16 64 bytes from 192.168.1.10: seq=33 ttl=61 time=2.240 ms
06:34:17 64 bytes from 192.168.1.10: seq=34 ttl=61 time=2.010 ms
06:34:18 64 bytes from 192.168.1.10: seq=35 ttl=61 time=2.290 ms
06:34:19 64 bytes from 192.168.1.10: seq=36 ttl=61 time=1.941 ms
06:34:20 64 bytes from 192.168.1.10: seq=37 ttl=61 time=2.452 ms
```

06:34:21 64 bytes from 192.168.1.10: seq=38 ttl=61 time=1.793 ms
06:34:22 64 bytes from 192.168.1.10: seq=39 ttl=61 time=2.350 ms
06:34:23 64 bytes from 192.168.1.10: seq=40 ttl=61 time=2.207 ms
06:34:24 64 bytes from 192.168.1.10: seq=41 ttl=61 time=2.233 ms
06:34:25 64 bytes from 192.168.1.10: seq=42 ttl=61 time=2.099 ms
06:34:26 64 bytes from 192.168.1.10: seq=43 ttl=61 time=2.659 ms
06:34:27 64 bytes from 192.168.1.10: seq=44 ttl=61 time=2.054 ms
06:34:28 64 bytes from 192.168.1.10: seq=45 ttl=61 time=2.289 ms
06:34:29 64 bytes from 192.168.1.10: seq=46 ttl=61 time=3.278 ms
06:34:30 64 bytes from 192.168.1.10: seq=47 ttl=61 time=2.008 ms
06:34:31 64 bytes from 192.168.1.10: seq=48 ttl=61 time=2.141 ms
06:34:32 64 bytes from 192.168.1.10: seq=49 ttl=61 time=2.152 ms
06:34:33 64 bytes from 192.168.1.10: seq=50 ttl=61 time=1.828 ms
06:34:34 64 bytes from 192.168.1.10: seq=51 ttl=61 time=1.955 ms
06:34:35 64 bytes from 192.168.1.10: seq=52 ttl=61 time=2.157 ms
06:34:36 64 bytes from 192.168.1.10: seq=53 ttl=61 time=2.103 ms
06:34:37 64 bytes from 192.168.1.10: seq=54 ttl=61 time=2.828 ms
06:34:38 64 bytes from 192.168.1.10: seq=55 ttl=61 time=2.295 ms
06:34:39 64 bytes from 192.168.1.10: seq=56 ttl=61 time=2.043 ms
06:34:40 64 bytes from 192.168.1.10: seq=57 ttl=61 time=2.095 ms
06:34:41 64 bytes from 192.168.1.10: seq=58 ttl=61 time=1.978 ms
06:34:42 64 bytes from 192.168.1.10: seq=59 ttl=61 time=1.930 ms
06:34:43 64 bytes from 192.168.1.10: seq=60 ttl=61 time=1.954 ms
06:34:44 64 bytes from 192.168.1.10: seq=61 ttl=61 time=2.090 ms
06:34:45 64 bytes from 192.168.1.10: seq=62 ttl=61 time=2.766 ms
06:34:46 64 bytes from 192.168.1.10: seq=63 ttl=61 time=2.359 ms
06:34:47 64 bytes from 192.168.1.10: seq=64 ttl=61 time=2.417 ms
06:34:48 64 bytes from 192.168.1.10: seq=65 ttl=61 time=2.599 ms
06:34:49 64 bytes from 192.168.1.10: seq=66 ttl=61 time=2.018 ms
06:34:50 64 bytes from 192.168.1.10: seq=67 ttl=61 time=2.284 ms
06:34:51 64 bytes from 192.168.1.10: seq=68 ttl=61 time=2.499 ms
06:34:52 64 bytes from 192.168.1.10: seq=69 ttl=61 time=1.945 ms
06:34:53 64 bytes from 192.168.1.10: seq=70 ttl=61 time=1.989 ms
06:34:54 64 bytes from 192.168.1.10: seq=71 ttl=61 time=2.025 ms
06:34:55 64 bytes from 192.168.1.10: seq=72 ttl=61 time=1.893 ms
06:34:56 64 bytes from 192.168.1.10: seq=73 ttl=61 time=2.510 ms
06:34:57 64 bytes from 192.168.1.10: seq=74 ttl=61 time=2.017 ms
06:34:58 64 bytes from 192.168.1.10: seq=75 ttl=61 time=2.120 ms
06:34:59 64 bytes from 192.168.1.10: seq=76 ttl=61 time=2.263 ms
06:35:00 64 bytes from 192.168.1.10: seq=77 ttl=61 time=2.216 ms

06:35:01 64 bytes from 192.168.1.10: seq=78 ttl=61 time=2.431 ms
06:35:02 64 bytes from 192.168.1.10: seq=79 ttl=61 time=1.984 ms

---

host 1 - 192.168.1.10 traceroute

/ # traceroute 192.168.1.10
traceroute to 192.168.1.10 (192.168.1.10), 30 hops max, 46 byte packets
1 192.168.4.1 (192.168.4.1) 0.692 ms 0.766 ms 0.349 ms
2 10.0.0.9 (10.0.0.9) 1.827 ms 1.573 ms 2.218 ms
3 10.0.0.1 (10.0.0.1) 2.420 ms 2.477 ms 2.167 ms
4 192.168.1.10 (192.168.1.10) 1.668 ms 3.237 ms 1.720 ms

R4's log: at 06:34:03 `eth1: NIC Link is Down`, then neighbor 3.3.3.3 went `Full -> Deleted`. R4 felt the broken backup link right away.

- **R1 route:** 192.168.4.0/24 is still via **10.0.0.2** (R2), cost **21**. The route age is 00:15:16, the same as the other routes. It was never recalculated, so the users' path never changed.
-
- **10.0.0.12/30** is gone from R1's table. That is the broken backup link, as I predicted.
-
- **Traceroute:** still the top path (10.0.0.9, then 10.0.0.1).
- Sequence numbers run **0 to 79 with no missing number**, and the clock times go up one second at a time.
- At **06:34:03** (seq 20), the same second R4's log says `NIC Link is Down`, the ping has a normal reply. The next lines are normal too.
- So the result is **0 seconds of outage, 0 pings lost**.

RESULT: outage 0 seconds, 0 pings lost (seq 0 to 79 continuous).
Users saw nothing, but redundancy is lost: the network now has no backup path.

---

test 5 primary and secondary path are gone

"Pasted image 20261103162028.png" could not be found.

test 5

----Host 1 to 192.168.1.10 ping ---

/ # ping 192.168.1.10 | while read l; do echo "$(date +%T) $l"; done
08:09:12 PING 192.168.1.10 (192.168.1.10): 56 data bytes
08:09:12 64 bytes from 192.168.1.10: seq=0 ttl=61 time=2.324 ms
08:09:13 64 bytes from 192.168.1.10: seq=1 ttl=61 time=1.961 ms
08:09:14 64 bytes from 192.168.1.10: seq=2 ttl=61 time=2.168 ms
08:09:15 64 bytes from 192.168.1.10: seq=3 ttl=61 time=2.219 ms
08:09:16 64 bytes from 192.168.1.10: seq=4 ttl=61 time=2.230 ms
08:09:17 64 bytes from 192.168.1.10: seq=5 ttl=61 time=1.982 ms
08:09:18 64 bytes from 192.168.1.10: seq=6 ttl=61 time=2.239 ms
08:09:19 64 bytes from 192.168.1.10: seq=7 ttl=61 time=2.148 ms
08:09:20 64 bytes from 192.168.1.10: seq=8 ttl=61 time=2.659 ms
08:09:21 64 bytes from 192.168.1.10: seq=9 ttl=61 time=1.937 ms
08:09:22 64 bytes from 192.168.1.10: seq=10 ttl=61 time=1.809 ms
08:09:23 64 bytes from 192.168.1.10: seq=11 ttl=61 time=1.963 ms
08:09:24 64 bytes from 192.168.1.10: seq=12 ttl=61 time=2.318 ms
08:09:25 64 bytes from 192.168.1.10: seq=13 ttl=61 time=2.574 ms
08:09:26 64 bytes from 192.168.1.10: seq=14 ttl=61 time=2.004 ms
08:09:27 64 bytes from 192.168.1.10: seq=15 ttl=61 time=1.887 ms
08:09:28 64 bytes from 192.168.1.10: seq=16 ttl=61 time=2.004 ms
08:09:29 64 bytes from 192.168.1.10: seq=17 ttl=61 time=1.905 ms
08:09:30 64 bytes from 192.168.1.10: seq=18 ttl=61 time=2.137 ms
08:09:31 64 bytes from 192.168.1.10: seq=19 ttl=61 time=2.351 ms
08:09:32 64 bytes from 192.168.1.10: seq=20 ttl=61 time=2.059 ms
08:09:33 64 bytes from 192.168.1.10: seq=21 ttl=61 time=2.166 ms
08:09:34 64 bytes from 192.168.1.10: seq=22 ttl=61 time=1.873 ms

--- R1's log ---

vyos@R1:~$ monitor log
Nov 03 08:03:24 vyos-netlinkd[2375]: RTM_NEWLINK -> eth0, state=DOWN, mac=0c:49:97:01:00:00
Nov 03 08:03:24 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on eth0:10.0.0.1: Full -> Deleted (KillNbr)
Nov 03 08:04:12 rsyslogd[1978]: -- MARK --
Nov 03 08:08:43 kernel: e1000: eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
Nov 03 08:08:43 vyos-netlinkd[2375]: RTM_NEWLINK -> eth0, state=UP, mac=0c:49:97:01:00:00
Nov 03 08:08:44 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on eth0:10.0.0.1: Down -> Init (HelloReceived)
Nov 03 08:08:53 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on eth0:10.0.0.1: Init -> ExStart (2-WayReceived)

Nov 03 08:08:53 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on eth0:10.0.0.1: ExStart -> Exchange (NegotiationDone)
Nov 03 08:08:53 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on eth0:10.0.0.1: Exchange -> Loading (ExchangeDone)
Nov 03 08:08:53 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on eth0:10.0.0.1: Loading -> Full (LoadingDone)
Nov 03 08:09:34 kernel: e1000: eth0 NIC Link is Down
Nov 03 08:09:34 vyos-netlinkd[2375]: RTM_NEWLINK -> eth0, state=DOWN, mac=0c:49:97:01:00:00
Nov 03 08:09:34 ospfd[1374]: AdjChg: Nbr 2.2.2.2, NbrIP 10.0.0.2 (default) on eth0:10.0.0.1: Full -> Deleted (KillNbr)

--- R1's ospf routes ---
vyos@R1:~$vyos@R1: show ip route ospf
Codes: K - kernel route, C - connected, L - local, S - static,
R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric, t - Table-Direct,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

IPv4 unicast VRF default:
O>* 3.3.3.3/32 [110/20] via 10.0.0.6, eth1, weight 1, 02:50:09

---- Host 1 Traceroute --

/ # traceroute 192.168.1.10
traceroute to 192.168.1.10 (192.168.1.10), 30 hops max, 46 byte packets
1 192.168.4.1 (192.168.4.1) 0.779 ms !N 0.532 ms !N 0.714 ms !N

Ping: the last reply is 08:09:34 (seq 22), and nothing comes after it. The R1 log shows eth0: NIC Link is Down at the same second, 08:09:34.
R1 route: only one OSPF route is left, 3.3.3.3/32 via R3. The route to 192.168.4.0/24 is gone. R1 can still reach R3, but R3 has no way to R4, so it is a dead end.
Traceroute: hop 1 is R4 (192.168.4.1), and it answers !N, which means “network unreachable”. R4 itself says it has no route back to the R1-side LAN. I said “stars or unreachable” earlier. It was unreachable.
The older log lines: 08:03:24 down and 08:08:43 up are from an earlier attempt, when you suspended and resumed R1-R2 before starting the ping. Your real test run is the last three lines at 08:09:34.

Test	Failure	Detection time	Seconds of outage	Path after	Failback time
5	Suspend R1-R2 and R3-R4	instant (link down)	**no end, until repaired**	none	(next step)

----- recovery data -------
turning on both suspended links to measure how long the network takes to heal after both links are suspended

Test	Failure	Detection	Outage	Path after	Failback
1	Suspend R1-R2	instant (link down)	about 3 s (2 pings lost)	R1-R3-R4	see Test 2
2	Resume R1-R2	n/a	0 s	R1-R2-R4	about 10 s
3	Stop R2	Dead timer (about 34 s)	about 34 s (33 pings lost)	R1-R3-R4	not tested
4	Suspend R3-R4	instant (link down)	0 s	R1-R2-R4 (unchanged)	not tested
5	Suspend R1-R2 and R3-R4	instant (link down)	until repaired	none	about 7 s

The main finding: losing a cable is detected in 3 seconds, but losing a router takes 34 seconds. That 34 seconds is what

---

PHASE 4 setting up BFD (Bidirectional Forwarding Decision)

- **Bidirectional:** both routers send and check, so each one confirms the other is alive.
- **Forwarding:** it tests the path that actually carries traffic, not just whether the cable looks connected.
- **Detection:** that is all it does. It doesn’t calculate routes.

What do i want to achieve ?

i want bfd to notify ospf to change path quick after a path or a neighbor dies

how it works ?

1. The two routers agree on a session: how often to send (e.g. every 300 ms) and how many misses are allowed (the multiplier, e.g. 3).
2. They exchange tiny, lightweight packets constantly.
3. If 3 in a row are missed, which takes about 900 ms here, BFD declares the neighbor down.
4. BFD tells OSPF right away. OSPF drops the neighbor immediately instead of waiting 40 seconds, recalculates, and traffic moves to the backup path.

BFD on the R1-R2 link only

R1
`set protocols bfd peer 10.0.0.2 source address 10.0.0.1`

`set protocols bfd peer 10.0.0.2 interval receive 300`

`set protocols bfd peer 10.0.0.2 interval transmit 300`

`set protocols bfd peer 10.0.0.2 interval multiplier 3`
` set protocols ospf interface eth0 bfd`

R2
`set protocols bfd peer 10.0.0.1 source address 10.0.0.2`

`set protocols bfd peer 10.0.0.1 interval receive 300`

`set protocols bfd peer 10.0.0.1 interval transmit 300`

`set protocols bfd peer 10.0.0.1 interval multiplier 3`

R1 - R3

R1
set protocols bfd peer 10.0.0.6 source address 10.0.0.5

set protocols bfd peer 10.0.0.6 interval receive 300

set protocols bfd peer 10.0.0.6 interval transmit 300

set protocols bfd peer 10.0.0.6 interval multiplier 3

set protocols ospf interface eth1 bfd

R3
set protocols bfd peer 10.0.0.5 source address 10.0.0.6

set protocols bfd peer 10.0.0.5 interval receive 300

set protocols bfd peer 10.0.0.5 interval transmit 300

set protocols bfd peer 10.0.0.5 interval multiplier 3

set protocols ospf interface eth0 bfd

R1- R2

set protocols bfd peer 10.0.0.2 source address 10.0.0.1

set protocols bfd peer 10.0.0.2 interval receive 300

set protocols bfd peer 10.0.0.2 interval transmit 300

set protocols bfd peer 10.0.0.2 interval multiplier 3

set protocols ospf interface eth0 bfd

R2

set protocols bfd peer 10.0.0.1 source address 10.0.0.2

set protocols bfd peer 10.0.0.1interval receive 300

set protocols bfd peer 10.0.0.1interval transmit 300

set protocols bfd peer 10.0.0.1 interval multiplier 3

set protocols ospf interface eth0 bfd

R2 - R4

R2

set protocols bfd peer 10.0.0.10 source address 10.0.0.9

set protocols bfd peer 10.0.0.10 interval receive 300

set protocols bfd peer 10.0.0.10 interval transmit 300

set protocols bfd peer 10.0.0.10 interval multiplier 3

set protocols ospf interface eth1 bfd

R4

set protocols bfd peer 10.0.0.9 source address 10.0.0.10

set protocols bfd peer 10.0.0.9 interval receive 300

set protocols bfd peer 10.0.0.9 interval transmit 300

set protocols bfd peer 10.0.0.9 interval multiplier 3

set protocols ospf interface eth0 bfd

R3- R4
R3
set protocols bfd peer 10.0.0.14 source address 10.0.0.13

set protocols bfd peer 10.0.0.14 interval receive 300

set protocols bfd peer 10.0.0.14 interval transmit 300

set protocols bfd peer 10.0.0.14 interval multiplier 3

set protocols ospf interface eth1 bfd

R4

set protocols bfd peer 10.0.0.13 source address 10.0.0.14

set protocols bfd peer 10.0.0.13 interval receive 300

set protocols bfd peer 10.0.0.13 interval transmit 300

set protocols bfd peer 10.0.0.13 interval multiplier 3

set protocols ospf interface eth1 bfd

Verification per router

Command used :
`show bfd peers`
`show ip ospf neighbor`
`show configuration commands | match ospf`

R1 output
vyos@R1:~$ show bfd peers
BFD Peers:
peer 10.0.0.6 local-address 10.0.0.5 vrf default interface eth1
ID: 1126738902
Remote ID: 1491804816
Active mode
Status: up
Uptime: 7 minute(s), 51 second(s)
Diagnostics: ok
Remote diagnostics: ok

Peer Type: configured
RTT min/avg/max: 0/0/0 usec
Local timers:
Detect-multiplier: 3
Receive interval: 300ms
Transmission interval: 300ms
Transmission interval (actual with jitter): 291ms
Detection timeout: 900ms
Echo receive interval: 50ms
Echo transmission interval: disabled
Remote timers:
Detect-multiplier: 3
Receive interval: 300ms
Transmission interval: 300ms
Echo receive interval: 50ms

```
peer 10.0.0.2 local-address 10.0.0.1 vrf default interface eth0
        ID: 1373924350
        Remote ID: 3868950796
        Active mode
        Status: up
        Uptime: 43 minute(s), 14 second(s)
        Diagnostics: ok
        Remote diagnostics: ok
        Peer Type: configured
        RTT min/avg/max: 0/0/0 usec
        Local timers:
                Detect-multiplier: 3
                Receive interval: 300ms
                Transmission interval: 300ms
                Transmission interval (actual with jitter): 294ms
                Detection timeout: 900ms
                Echo receive interval: 50ms
                Echo transmission interval: disabled
        Remote timers:
                Detect-multiplier: 3
                Receive interval: 300ms
                Transmission interval: 300ms
                Echo receive interval: 50ms
```

vyos@R1:~$ show ip ospf neighbor

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
2.2.2.2 1 Full/- 1h06m33s 37.280s 10.0.0.2 eth0:10.0.0.1 0 0 0
3.3.3.3 1 Full/- 1h06m23s 38.470s 10.0.0.6 eth1:10.0.0.5 0 0 0

vyos@R1:~$ show configuration commands | match ospf
set protocols ospf interface eth0 area '0'
set protocols ospf interface eth0 bfd
set protocols ospf interface eth0 cost '10'
set protocols ospf interface eth0 network 'point-to-point'
set protocols ospf interface eth1 area '0'
set protocols ospf interface eth1 bfd
set protocols ospf interface eth1 cost '20'
set protocols ospf interface eth1 network 'point-to-point'
set protocols ospf interface eth2 area '0'
set protocols ospf interface eth2 network 'point-to-point'
set protocols ospf interface eth2 passive
set protocols ospf interface lo area '0'
set protocols ospf log-adjacency-changes detail
set protocols ospf parameters router-id '1.1.1.1'

R2 output
vyos@R2:~$ show bfd peers
BFD Peers:
peer 10.0.0.10 local-address 10.0.0.9 vrf default interface eth1
ID: 937598730
Remote ID: 961031016
Active mode
Status: up
Uptime: 21 minute(s), 21 second(s)
Diagnostics: ok
Remote diagnostics: ok
Peer Type: configured
RTT min/avg/max: 0/0/0 usec
Local timers:
Detect-multiplier: 3
Receive interval: 300ms
Transmission interval: 300ms
Transmission interval (actual with jitter): 252ms
Detection timeout: 900ms
Echo receive interval: 50ms
Echo transmission interval: disabled

Remote timers:
Detect-multiplier: 3
Receive interval: 300ms
Transmission interval: 300ms
Echo receive interval: 50ms

```
     peer 10.0.0.1 local-address 10.0.0.2 vrf default
             ID: 3868950796
             Remote ID: 1373924350
             Active mode
             Status: up
             Uptime: 31 minute(s), 8 second(s)
             Diagnostics: ok
             Remote diagnostics: ok
             Peer Type: configured
             RTT min/avg/max: 0/0/0 usec
             Local timers:
                     Detect-multiplier: 3
                     Receive interval: 300ms
                     Transmission interval: 300ms
                     Transmission interval (actual with jitter): 273ms
                     Detection timeout: 900ms
                     Echo receive interval: 50ms
                     Echo transmission interval: disabled
             Remote timers:
                     Detect-multiplier: 3
                     Receive interval: 300ms
                     Transmission interval: 300ms
                     Echo receive interval: 50ms
```

vyos@R2:~$ show ip ospf neighbor

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
1.1.1.1 1 Full/- 1h04m09s 32.363s 10.0.0.1 eth0:10.0.0.2 0 0 0
4.4.4.4 1 Full/- 1h04m08s 33.388s 10.0.0.10 eth1:10.0.0.9 0 0 0

vyos@R2:~$ show configuration commands | match ospf
set protocols ospf interface eth0 area '0'
set protocols ospf interface eth0 cost '10'
set protocols ospf interface eth0 network 'point-to-point'
set protocols ospf interface eth1 area '0'
set protocols ospf interface eth1 bfd
set protocols ospf interface eth1 cost '10'
set protocols ospf interface eth1 network 'point-to-point'

set protocols ospf interface lo area '0'
set protocols ospf log-adjacency-changes detail
set protocols ospf parameters router-id '2.2.2.2'

R3 output
vyos@R3:~$ show bfd peers
BFD Peers:
peer 10.0.0.14 local-address 10.0.0.13 vrf default interface eth1
ID: 3732792203
Remote ID: 858983013
Active mode
Status: up
Uptime: 32 minute(s), 44 second(s)
Diagnostics: ok
Remote diagnostics: ok
Peer Type: configured
RTT min/avg/max: 0/0/0 usec
Local timers:
Detect-multiplier: 3
Receive interval: 300ms
Transmission interval: 300ms
Transmission interval (actual with jitter): 267ms
Detection timeout: 900ms
Echo receive interval: 50ms
Echo transmission interval: disabled
Remote timers:
Detect-multiplier: 3
Receive interval: 300ms
Transmission interval: 300ms
Echo receive interval: 50ms

```
peer 10.0.0.5 local-address 10.0.0.6 vrf default interface eth0
        ID: 1491804816
        Remote ID: 1126738902
        Active mode
        Status: up
        Uptime: 12 minute(s), 44 second(s)
        Diagnostics: ok
        Remote diagnostics: ok
        Peer Type: configured
        RTT min/avg/max: 0/0/0 usec
        Local timers:
```

```
                Detect-multiplier: 3
                Receive interval: 300ms
                Transmission interval: 300ms
                Transmission interval (actual with jitter): 225ms
                Detection timeout: 900ms
                Echo receive interval: 50ms
                Echo transmission interval: disabled
        Remote timers:
                Detect-multiplier: 3
                Receive interval: 300ms
                Transmission interval: 300ms
                Echo receive interval: 50ms
```

vyos@R3:~$ show configuration commands | match ospf
set protocols ospf interface eth0 area '0'
set protocols ospf interface eth0 bfd
set protocols ospf interface eth0 cost '20'
set protocols ospf interface eth0 network 'point-to-point'
set protocols ospf interface eth1 area '0'
set protocols ospf interface eth1 bfd
set protocols ospf interface eth1 cost '20'
set protocols ospf interface eth1 network 'point-to-point'
set protocols ospf interface lo area '0'
set protocols ospf log-adjacency-changes detail
set protocols ospf parameters router-id '3.3.3.3'

R4 output
vyos@R4:~$ show bfd peers
BFD Peers:
peer 10.0.0.9 local-address 10.0.0.10 vrf default interface eth0
ID: 961031016
Remote ID: 937598730
Active mode
Status: up
Uptime: 6 minute(s), 10 second(s)
Diagnostics: ok
Remote diagnostics: ok
Peer Type: configured
RTT min/avg/max: 0/0/0 usec
Local timers:
Detect-multiplier: 3
Receive interval: 300ms

Transmission interval: 300ms
Transmission interval (actual with jitter): 225ms
Detection timeout: 900ms
Echo receive interval: 50ms
Echo transmission interval: disabled
Remote timers:
Detect-multiplier: 3
Receive interval: 300ms
Transmission interval: 300ms
Echo receive interval: 50ms

```
    peer 10.0.0.13 local-address 10.0.0.14 vrf default interface eth1
            ID: 858983013
            Remote ID: 3732792203
            Active mode
            Status: up
            Uptime: 34 second(s)
            Diagnostics: ok
            Remote diagnostics: ok
            Peer Type: configured
            RTT min/avg/max: 0/0/0 usec
            Local timers:
                    Detect-multiplier: 3
                    Receive interval: 300ms
                    Transmission interval: 300ms
                    Transmission interval (actual with jitter): 282ms
                    Detection timeout: 900ms
                    Echo receive interval: 50ms
                    Echo transmission interval: disabled
            Remote timers:
                    Detect-multiplier: 3
                    Receive interval: 300ms
                    Transmission interval: 300ms
                    Echo receive interval: 50ms
```

vyos@R4:~$ show ip ospf neighbor

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
2.2.2.2 1 Full/- 39m00s 38.723s 10.0.0.9 eth0:10.0.0.10 0 0 0
3.3.3.3 1 Full/- 38m59s 39.898s 10.0.0.13 eth1:10.0.0.14 0 0 0

vyos@R4:~$ show configuration commands | match ospf
set protocols ospf interface eth0 area '0'
set protocols ospf interface eth0 bfd

set protocols ospf interface eth0 cost '10'
set protocols ospf interface eth0 network 'point-to-point'
set protocols ospf interface eth1 area '0'
set protocols ospf interface eth1 bfd
set protocols ospf interface eth1 cost '20'
set protocols ospf interface eth1 network 'point-to-point'
set protocols ospf interface eth2 area '0'
set protocols ospf interface eth2 network 'point-to-point'
set protocols ospf interface eth2 passive
set protocols ospf interface lo area '0'
set protocols ospf log-adjacency-changes detail
set protocols ospf parameters router-id '4.4.4.4'
vyos@R4:~$

Phase 4 conclusion

To see the effect of BFD, I repeated the same test as before: I ran a continuous ping and stopped R2 in GNS3. Before BFD, I lost 33 pings (about 34 seconds). With BFD configured (300 ms interval, multiplier 3, so 900 ms detection time), I lost 0 pings. BFD noticed that R2 died in under 1 second and told OSPF, and OSPF moved the traffic to the R1-R3-R4 path.

Trouble shooting tickets

Ticket 1 (BFD alert on R4)

```
**INC-001, low priority.** Monitoring shows a BFD peer on R4 with a bad
status. Users aren't affected yet. Find which peer, find why, and fix it.
```

My procedure :

1st: reading the ticket carefully it gives me a hint that there might be a problem with the peers of r4 along the low priority path which is the backup path (r1-r3-r4) , so ill check the r4 first and then the r3 second to confirm my initial diagnosis.

ipon checking

```
vyos@R4:~$ show bfd peers
BFD Peers:
        peer 10.0.0.12 local-address 10.0.0.14 vrf default
                ID: 275787142
                Remote ID: 0
```

```
                Active mode
                ******Status: down******
                Downtime: 7 minute(s), 55 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 780ms
                        Detection timeout: 3000ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 1000ms
                        Transmission interval: 1000ms
                        Echo receive interval: disabled
```

the status down gives another hint that there might be wrong between the link of r3 - r4

I'll investigate it with the command:

```
show ip ospf neighbor
```

as i check both r4's neighbor are up which means that the problem must be the BFD configurations there's a possibility that a undocumented change happened

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
2.2.2.2 1 Full/- 11m23s 36.608s 10.0.0.9 eth0:10.0.0.10 0 0 0
3.3.3.3 1 Full/- 11m22s 37.700s 10.0.0.13 eth1:10.0.0.14 0 0 0

so i cross checked the commands responsible for BFD peering and i see it

R3's config
os@R3:~$ show configuration commands | match bfd
set protocols bfd peer 10.0.0.5 interval multiplier '3'
set protocols bfd peer 10.0.0.5 interval receive '300'
set protocols bfd peer 10.0.0.5 interval transmit '300'
set protocols bfd peer 10.0.0.5 source address '10.0.0.6'
set protocols bfd peer 10.0.0.14 interval multiplier '3'

set protocols bfd peer 10.0.0.14 interval receive '300'
set protocols bfd peer 10.0.0.14 interval transmit '300'
set protocols bfd peer 10.0.0.14 source address '10.0.0.13'
set protocols ospf interface eth0 bfd
set protocols ospf interface eth1 bfd

this orange mark is the command that states that 10.0.0.14 is the neighbors ip and 10.0.0.13 is R3's ip on that link but somehow R4's configuration is this
R4's config

vyos@R4:~$ show configuration commands | match bfd
set protocols bfd peer 10.0.0.9 interval multiplier '3'
set protocols bfd peer 10.0.0.9 interval receive '300'
set protocols bfd peer 10.0.0.9 interval transmit '300'
set protocols bfd peer 10.0.0.9 source address '10.0.0.10'
set protocols bfd peer 10.0.0.12 interval multiplier '3'
set protocols bfd peer 10.0.0.12 interval receive '300'
set protocols bfd peer 10.0.0.12 interval transmit '300'
set protocols bfd peer 10.0.0.12 source address '10.0.0.14'
set protocols ospf interface eth0 bfd
set protocols ospf interface eth1 bfd

it uses the its network address instead of the neighbor's ip address which is the ip of R3 on that link

so the main problem here is the wrong ip address used in the bfd commands

---

Ticket 2

```
**medium priority.** After last night’s maintenance, users report a long
outage whenever a router in the primary path goes down. The monitoring
dashboard shows **every BFD session as up**. Find out why failover is slow and
fix it.
```

/ # ping 192.168.4.10
PING 192.168.4.10 (192.168.4.10): 56 data bytes
64 bytes from 192.168.4.10: seq=0 ttl=61 time=2.708 ms
64 bytes from 192.168.4.10: seq=1 ttl=61 time=3.338 ms
64 bytes from 192.168.4.10: seq=2 ttl=61 time=33.678 ms
64 bytes from 192.168.4.10: seq=3 ttl=61 time=66.935 ms
64 bytes from 192.168.4.10: seq=4 ttl=61 time=47.107 ms

64 bytes from 192.168.4.10: seq=5 ttl=61 time=104.889 ms
64 bytes from 192.168.4.10: seq=6 ttl=61 time=13.993 ms
64 bytes from 192.168.4.10: seq=7 ttl=61 time=15.926 ms
64 bytes from 192.168.4.10: seq=8 ttl=61 time=163.895 ms
64 bytes from 192.168.4.10: seq=9 ttl=61 time=24.045 ms
64 bytes from 192.168.4.10: seq=10 ttl=61 time=94.588 ms
64 bytes from 192.168.4.10: seq=11 ttl=61 time=62.408 ms
64 bytes from 192.168.4.10: seq=12 ttl=61 time=54.318 ms
64 bytes from 192.168.4.10: seq=13 ttl=61 time=55.743 ms
64 bytes from 192.168.4.10: seq=14 ttl=61 time=80.489 ms
64 bytes from 192.168.4.10: seq=15 ttl=61 time=28.911 ms
64 bytes from 192.168.4.10: seq=16 ttl=61 time=82.371 ms
64 bytes from 192.168.4.10: seq=19 ttl=61 time=226.662 ms
64 bytes from 192.168.4.10: seq=20 ttl=61 time=8.555 ms
64 bytes from 192.168.4.10: seq=21 ttl=61 time=11.448 ms
64 bytes from 192.168.4.10: seq=22 ttl=61 time=47.781 ms
64 bytes from 192.168.4.10: seq=23 ttl=61 time=45.385 ms
64 bytes from 192.168.4.10: seq=24 ttl=61 time=55.441 ms

Actions taken :

1st I'll be checking to see the bfd peering just to verify the monitoring dashboard is truly up
: all is up
R1
vyos@R1:~$ show bfd peers
BFD Peers:
peer 10.0.0.6 local-address 10.0.0.5 vrf default
ID: 635603548
Remote ID: 3553035295
Active mode
Status: up

peer 10.0.0.2 local-address 10.0.0.1 vrf default
ID: 1166526374
Remote ID: 2682123991
Active mode
Status: up

R2
vyos@R2:~$ show bfd peers
BFD Peers:
peer 10.0.0.1 local-address 10.0.0.2 vrf default

ID: 2682123991
Remote ID: 1166526374
Active mode
Status: up

peer 10.0.0.10 local-address 10.0.0.9 vrf default
ID: 891841461
Remote ID: 4221161167
Active mode
Status: up

R3

vyos@R3:~$ show bfd peers
BFD Peers:
peer 10.0.0.5 local-address 10.0.0.6 vrf default
ID: 3553035295
Remote ID: 635603548
Active mode
Status: up
peer 10.0.0.14 local-address 10.0.0.13 vrf default
ID: 2203500475
Remote ID: 1583661678
Active mode
Status: up

R4
vyos@R4:~$ show bfd peers
BFD Peers:
peer 10.0.0.13 local-address 10.0.0.14 vrf default
ID: 1583661678
Remote ID: 2203500475
Active mode
Status: up

```
peer 10.0.0.9 local-address 10.0.0.10 vrf default
            ID: 4221161167
            Remote ID: 891841461
            Active mode
            Status: up
            Uptime: 24 minute(s), 2 second(s)
```

2nd ill make sure that the ospf neighbor is up as well

r1
vyos@R1:~$ show ip ospf nei

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
2.2.2.2 1 Full/- 25m02s 38.025s 10.0.0.2 eth0:10.0.0.1 0 0 0
3.3.3.3 1 Full/- 28m37s 31.195s 10.0.0.6 eth1:10.0.0.5 0 0 0

r2

vyos@R2:~$ show ip ospf nei

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
1.1.1.1 1 Full/- 25m09s 37.129s 10.0.0.1 eth0:10.0.0.2 0 0 0
4.4.4.4 1 Full/- 25m13s 36.817s 10.0.0.10 eth1:10.0.0.9 0 0 0

r3

vyos@R3:~$ show ip ospf nei

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
1.1.1.1 1 Full/- 28m54s 36.441s 10.0.0.5 eth0:10.0.0.6 0 0 0
4.4.4.4 1 Full/- 28m54s 36.149s 10.0.0.14 eth1:10.0.0.13 0 0 0

r4

vyos@R4:~$ show ip ospf neighbor

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
2.2.2.2 1 Full/- 25m30s 34.918s 10.0.0.9 eth0:10.0.0.10 0 0 0
3.3.3.3 1 Full/- 29m00s 38.103s 10.0.0.13 eth1:10.0.0.14 0 0 0

---

then ill check the bfd configuration for each router

As it turns out router 1 is missing a command that tells the router to use bfd on eth0 interface to R2 so that when the router2 goes down the failover path should immediately takeover because it will no longer use the default 40 seconds dead timer
vyos@R1:~$ show configuration commands | match bfd
set protocols bfd peer 10.0.0.2 interval multiplier '3'
set protocols bfd peer 10.0.0.2 interval receive '300'
set protocols bfd peer 10.0.0.2 interval transmit '300'
set protocols bfd peer 10.0.0.2 source address '10.0.0.1'

set protocols bfd peer 10.0.0.6 interval multiplier '3'
set protocols bfd peer 10.0.0.6 interval receive '300'
set protocols bfd peer 10.0.0.6 interval transmit '300'
set protocols bfd peer 10.0.0.6 source address '10.0.0.5'
set protocols ospf interface eth1 bfd

to fix this apply this
set protocols ospf interface eth0 bfd
so that ospf uses bfd to detect whether the neighbor is down or not

then test it again by running a ping then intentionally stopping R2

R1's ospf neigbor
vyos@R1# exit
exit
vyos@R1:~$ show ip ospf neigh

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
3.3.3.3 1 Full/- 59m15s 33.948s 10.0.0.6 eth1:10.0.0.5 0 0 0

/ # ping 192.168.4.10
PING 192.168.4.10 (192.168.4.10): 56 data bytes
64 bytes from 192.168.4.10: seq=0 ttl=61 time=3.954 ms
64 bytes from 192.168.4.10: seq=1 ttl=61 time=53.788 ms
64 bytes from 192.168.4.10: seq=2 ttl=61 time=79.442 ms
64 bytes from 192.168.4.10: seq=3 ttl=61 time=26.903 ms
64 bytes from 192.168.4.10: seq=4 ttl=61 time=17.141 ms
64 bytes from 192.168.4.10: seq=5 ttl=61 time=14.702 ms
64 bytes from 192.168.4.10: seq=6 ttl=61 time=92.909 ms
64 bytes from 192.168.4.10: seq=7 ttl=61 time=42.794 ms
64 bytes from 192.168.4.10: seq=8 ttl=61 time=43.184 ms
64 bytes from 192.168.4.10: seq=10 ttl=61 time=32.210 ms
64 bytes from 192.168.4.10: seq=11 ttl=61 time=32.968 ms
64 bytes from 192.168.4.10: seq=12 ttl=61 time=30.877 ms
64 bytes from 192.168.4.10: seq=13 ttl=61 time=91.517 ms
64 bytes from 192.168.4.10: seq=14 ttl=61 time=204.067 ms
64 bytes from 192.168.4.10: seq=15 ttl=61 time=135.816 ms
64 bytes from 192.168.4.10: seq=16 ttl=61 time=117.414 ms
64 bytes from 192.168.4.10: seq=17 ttl=61 time=13.715 ms
64 bytes from 192.168.4.10: seq=18 ttl=61 time=169.179 ms
64 bytes from 192.168.4.10: seq=19 ttl=61 time=8.946 ms
64 bytes from 192.168.4.10: seq=20 ttl=61 time=71.837 ms

64 bytes from 192.168.4.10: seq=21 ttl=61 time=52.765 ms
64 bytes from 192.168.4.10: seq=22 ttl=61 time=31.188 ms
64 bytes from 192.168.4.10: seq=23 ttl=61 time=216.090 ms
64 bytes from 192.168.4.10: seq=24 ttl=61 time=68.888 ms
64 bytes from 192.168.4.10: seq=25 ttl=61 time=65.506 ms
64 bytes from 192.168.4.10: seq=26 ttl=61 time=176.901 ms
64 bytes from 192.168.4.10: seq=27 ttl=61 time=165.395 ms
64 bytes from 192.168.4.10: seq=28 ttl=61 time=41.858 ms
^C
--- 192.168.4.10 ping statistics ---
29 packets transmitted, 28 packets received, 3% packet loss
round-trip min/avg/max = 3.954/75.069/216.090 ms

as shown here from two packets it went down to just one packet loss after the fix which means that bfd immediately notify ospf and ospf used the backup path as it was notified that R2 went down

Why the outage was short even with the fault: R4 also has BFD toward R2, so R4 noticed R2 died within 1 second. R4 then told the other routers, including R1, through OSPF, via R3. So R1 learned R2 was gone from R4, even though R1's own BFD wasn't wired up.

---

ticket 3
pre ticket checks

```
show bfd peers
```

R1:

```
vyos@R1:~$ show bfd peers
BFD Peers:
        peer 10.0.0.6 local-address 10.0.0.5 vrf default
                ID: 635603548
                Remote ID: 3553035295
                Active mode
                Status: up
                Uptime: 1 hour(s), 7 minute(s), 34 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
```

```
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 276ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms

        peer 10.0.0.2 local-address 10.0.0.1 vrf default
                ID: 1166526374
                Remote ID: 2374230101
                Active mode
                Status: up
                Uptime: 3 minute(s), 44 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 264ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms
```

R2

```
vyos@R2:~$ show bfd peers
BFD Peers:
        peer 10.0.0.1 local-address 10.0.0.2 vrf default
                ID: 2374230101
                Remote ID: 1166526374
                Active mode
```

```
                Status: up
                Uptime: 3 minute(s), 3 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 297ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms

        peer 10.0.0.10 local-address 10.0.0.9 vrf default
                ID: 967942224
                Remote ID: 4221161167
                Active mode
                Status: up
                Uptime: 3 minute(s), 3 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 261ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
:
                        Echo receive interval: 50ms

        peer 10.0.0.10 local-address 10.0.0.9 vrf default
                ID: 967942224
```

```
                Remote ID: 4221161167
                Active mode
                Status: up
                Uptime: 3 minute(s), 3 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 261ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms
```

R3

```
vyos@R3:~$ show bfd peers
BFD Peers:
        peer 10.0.0.5 local-address 10.0.0.6 vrf default
                ID: 3553035295
                Remote ID: 635603548
                Active mode
                Status: up
                Uptime: 1 hour(s), 7 minute(s), 40 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 300ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
```

```
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms

        peer 10.0.0.14 local-address 10.0.0.13 vrf default
                ID: 2203500475
                Remote ID: 1583661678
                Active mode
                Status: up
                Uptime: 1 hour(s), 7 minute(s), 40 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 261ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms
```

R4

```
vyos@R4:~$ show bfd peers
BFD Peers:
        peer 10.0.0.13 local-address 10.0.0.14 vrf default
                ID: 1583661678
                Remote ID: 2203500475
                Active mode
                Status: up
                Uptime: 1 hour(s), 7 minute(s), 46 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
```

```
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 261ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms


 peer 10.0.0.9 local-address 10.0.0.10 vrf default
                ID: 4221161167
                Remote ID: 967942224
                Active mode
                Status: up
                Uptime: 3 minute(s), 55 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 240ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms
```

INC-003, high priority. The NOC sees OSPF neighbors repeatedly dropping and coming back on their own, with no cable touched and no router rebooted. Find the cause and fix it permanently.

```
vyos@R3:~$ monitor log
Nov 07 06:38:57 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Exchange -> Full (ExchangeDone)
Nov 07 06:38:57 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Full -> Deleted (InactivityTimer)
Nov 07 06:38:57 bfdd[1363]: [EC 100663315] CPU starvation: {(event
*)0x7ffcd5df1e00 arg=0x556b19b92c40 ready               bfd_xmt_cb() &bs-
>xmttimer_ev from ../bfdd/event.c:149} getting executed 31ms late, warning
threshold 10ms. System load: 0.00, 0.00, 0.00
Nov 07 06:39:07 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Down -> Init (HelloReceived)
Nov 07 06:39:07 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Init -> ExStart (2-WayReceived)
Nov 07 06:39:07 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: ExStart -> Exchange (NegotiationDone)
Nov 07 06:39:07 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Exchange -> Full (ExchangeDone)
Nov 07 06:39:07 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Full -> Deleted (InactivityTimer)
Nov 07 06:39:08 ospfd[1380]: [EC 134217741] Link State Acknowledgment: Unknown
Neighbor 4.4.4.4
Nov 07 06:39:08 bfdd[1363]: [EC 100663315] CPU starvation: {(event
*)0x7ffcd5df1e00 arg=0x556b19b92c40 ready               bfd_xmt_cb() &bs-
>xmttimer_ev from ../bfdd/event.c:149} getting executed 11ms late, warning
threshold 10ms. System load: 0.08, 0.02, 0.01
Nov 07 06:39:17 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Down -> Init (HelloReceived)
Nov 07 06:39:17 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Init -> ExStart (2-WayReceived)
Nov 07 06:39:17 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: ExStart -> Exchange (NegotiationDone)
Nov 07 06:39:17 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Exchange -> Full (ExchangeDone)
Nov 07 06:39:17 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Full -> Deleted (InactivityTimer)
Nov 07 06:39:18 ospfd[1380]: [EC 134217741] Link State Acknowledgment: Unknown
Neighbor 4.4.4.4
Nov 07 06:39:19 bfdd[1363]: [EC 100663315] CPU starvation: {(event
*)0x7ffcd5df1e00 arg=0x556b19b92c40 ready               bfd_xmt_cb() &bs-
>xmttimer_ev from ../bfdd/event.c:149} getting executed 47ms late, warning
threshold 10ms. System load: 0.07, 0.02, 0.00
Nov 07 06:39:27 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Down -> Init (HelloReceived)
Nov 07 06:39:27 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Init -> ExStart (2-WayReceived)
Nov 07 06:39:27 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
```

```
eth1:10.0.0.13: ExStart -> Exchange (NegotiationDone)
Nov 07 06:39:27 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Exchange -> Full (ExchangeDone)
Nov 07 06:39:28 ospfd[1380]: AdjChg: Nbr 4.4.4.4, NbrIP 10.0.0.14 (default) on
eth1:10.0.0.13: Full -> Deleted (InactivityTimer)
Nov 07 06:39:31 bfdd[1363]: [EC 100663315] CPU starvation: {(event
*)0x7ffcd5df1e00 arg=0x556b19b92c40 ready               bfd_xmt_cb() &bs-
>xmttimer_ev from ../bfdd/event.c:149} getting executed 23ms late, warning
threshold 10ms. System load: 0.06, 0.01, 0.00
```

this monitor log confirms what the Other NOC said
the ospf neighbor keeps resetting

and both routers aren't appearing on both show ip ospf neighbor table
vyos@R3:~$ show ip ospf neighbor

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
1.1.1.1 1 Full/- 1h54m29s 33.104s 10.0.0.5 eth0:10.0.0.6 0 0 0

vyos@R4:~$ show ip ospf neighbor

Neighbor ID Pri State Up Time Dead Time Address Interface RXmtL RqstL DBsmL
2.2.2.2 1 Full/- 51m03s 36.967s 10.0.0.9 eth0:10.0.0.10 1 0 0

as per checking on the show bfd peers on both routers
both have the detection timeout is Detection timeout: 2000ms
which from the other healthy link is 900

BFD Peers:
peer 10.0.0.13 local-address 10.0.0.14 vrf default
ID: 1583661678
Remote ID: 2203500475
Active mode
Status: init
Diagnostics: control detection time expired
Remote diagnostics: control detection time expired
Peer Type: configured
RTT min/avg/max: 0/0/0 usec
Local timers:
Detect-multiplier: 2
Receive interval: 20ms
Transmission interval: 20ms
Transmission interval (actual with jitter): 870ms
Detection timeout: 2000ms

Echo receive interval: 50ms
Echo transmission interval: disabled
Remote timers:
Detect-multiplier: 2
Receive interval: 1000ms
Transmission interval: 1000ms
Echo receive interval: 50ms

r3
peer 10.0.0.14 local-address 10.0.0.13 vrf default
ID: 2203500475
Remote ID: 1583661678
Active mode
Status: init
Diagnostics: control detection time expired
Remote diagnostics: control detection time expired
Peer Type: configured
RTT min/avg/max: 0/0/0 usec
Local timers:
Detect-multiplier: 2
Receive interval: 20ms
Transmission interval: 20ms
Transmission interval (actual with jitter): 810ms
Detection timeout: 2000ms
Echo receive interval: 50ms
Echo transmission interval: disabled
Remote timers:
Detect-multiplier: 2
Receive interval: 1000ms
Transmission interval: 1000ms
Echo receive interval: 50ms

Both routers are configured to exchange BFD packets every 20 ms. This aggressive interval may be too sensitive for the virtual environment, where CPU scheduling or packet-processing delays can occur. If BFD packets aren't received within the negotiated detection timeout, BFD declares the session DOWN and notifies OSPF, causing the neighbor adjacency to drop. Once connectivity is considered available again, BFD and OSPF re-establish their sessions. If the same timeout happens repeatedly, the OSPF neighbor keeps flapping.
Repeated BFD state transitions can cause OSPF neighbor flapping.

to fix it

R4

```
configure
set protocols bfd peer 10.0.0.13 interval receive 300
set protocols bfd peer 10.0.0.13 interval transmit 300
set protocols bfd peer 10.0.0.13 interval multiplier 3
commit
```

R3

```
configure
set protocols bfd peer 10.0.0.14 interval receive 300
set protocols bfd peer 10.0.0.14 interval transmit 300
set protocols bfd peer 10.0.0.14 interval multiplier 3
commit
```

Command	Purpose
`interval receive 300`	Requests BFD packets every 300 ms or slower
`interval transmit 300`	Sends BFD packets every 300 ms
`interval multiplier 3`	Allows 3 negotiated intervals before declaring failure

After the fix was applied

```
vyos@R4:~$ show bfd peers
BFD Peers:
        peer 10.0.0.13 local-address 10.0.0.14 vrf default
                ID: 1583661678
                Remote ID: 2203500475
                Active mode
                Status: up
                Uptime: 3 minute(s), 40 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 279ms
                        Detection timeout: 900ms
```

```
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms
```

```
vyos@R4:~$ show ip ospf nei

Neighbor ID     Pri State           Up Time         Dead Time Address
Interface                        RXmtL RqstL DBsmL
2.2.2.2           1 Full/-          1h28m25s          35.707s 10.0.0.9
eth0:10.0.0.10                       0     0     0
3.3.3.3           1 Full/-          10m13s            34.478s 10.0.0.13
eth1:10.0.0.14                       0     0     0
```

vyos@R3:~$ show bfd peers

```
peer 10.0.0.14 local-address 10.0.0.13 vrf default
                ID: 2203500475
                Remote ID: 1583661678
                Active mode
                Status: up
                Uptime: 3 minute(s), 24 second(s)
                Diagnostics: ok
                Remote diagnostics: ok
                Peer Type: configured
                RTT min/avg/max: 0/0/0 usec
                Local timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Transmission interval (actual with jitter): 240ms
                        Detection timeout: 900ms
                        Echo receive interval: 50ms
                        Echo transmission interval: disabled
                Remote timers:
                        Detect-multiplier: 3
                        Receive interval: 300ms
                        Transmission interval: 300ms
                        Echo receive interval: 50ms
```

```
vyos@R3:~$ show ip ospf neighbor

Neighbor ID     Pri State           Up Time         Dead Time Address
Interface                       RXmtL RqstL DBsmL
1.1.1.1           1 Full/-          2h32m31s          31.510s 10.0.0.5
eth0:10.0.0.6                        0     0     0
4.4.4.4           1 Full/-          10m38s            31.796s 10.0.0.14
eth1:10.0.0.13                       0     0     0
```

Faster BFD timers are only safe if the device can send on time. Real routers handle 50 ms or lower because they have dedicated hardware, but virtual routers in GNS3 share a laptop’s CPU. That is why 300 ms × 3 is the sensible value for this lab.